

Polar Coordinates

MATH 146 – Calculus II

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Polar Coordinates

The Cartesian coordinate (x, y) is the foundation of our visual understanding of calculus, but there is also the **polar coordinate system** for the plane.

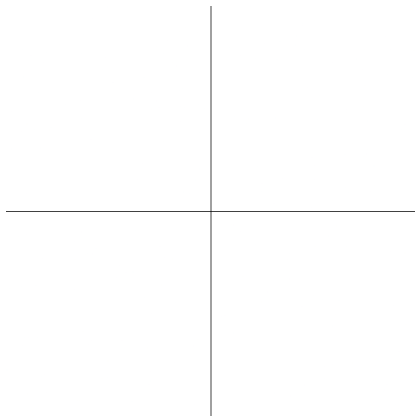
Polar Coordinates

Instead of defining a point using x and y , polar coordinates define a point using r , the straight line distance from the point to the origin, and θ , the angle between the line drawn from the origin to the point and the x -axis measured counter-clockwise.

Plotting Polar Coordinates

Example: Plot each of the following polar coordinate pairs on the axes below.

$$(1, \pi/3), \quad (2, \pi/3), \quad (2, 4\pi/3), \quad (-3, 4\pi/3), \quad (1, 7\pi/3), \quad (1, -\pi/3)$$



Converting Between Polar and Cartesian Coordinates

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Given a point P in the plane with Cartesian coordinates (x, y) and polar coordinates (r, θ) , the following conversion formulas hold true:

$$x = r \cos(\theta) \quad \text{and} \quad y = r \sin(\theta)$$

$$r^2 = x^2 + y^2 \quad \text{and} \quad \tan(\theta) = \frac{y}{x}$$

These formulas can be used to convert from rectangular to polar coordinates or from polar to rectangular coordinates.

Polar Functions

Polar Functions

A **polar function** expresses the radius r as a function of the angle θ .

Sometimes it is more natural to plot things in polar coordinates.

Example: Polar Functions

Example: Plot $f(\theta) = 3$.

Example: Plot $f(\theta) = \theta$.

Example: Polar Functions

Example: Plot $f(\theta) = 2 \sin(\theta)$.

Example: Plot $f(\theta) = 2 \cos(2\theta) \sec(\theta)$.

There are many elegant curves that can be plotted with polar coordinates but not Cartesian coordinates. (Try $3 \sin(\pi\theta)$, $2 - 2 \sin(\theta)$ and more.)

Polar Functions

These graphs are literally function graphs, and so we can apply calculus to the, (See you on Monday!) However, just as with Cartesian coordinates, you can also have graphs of polar equations.

Example: $\theta = \pi$.

Example: $3 \cos(\theta) - 2 \sin(\theta) = \frac{1}{r}$.